近红外多光谱传感器设计和进度

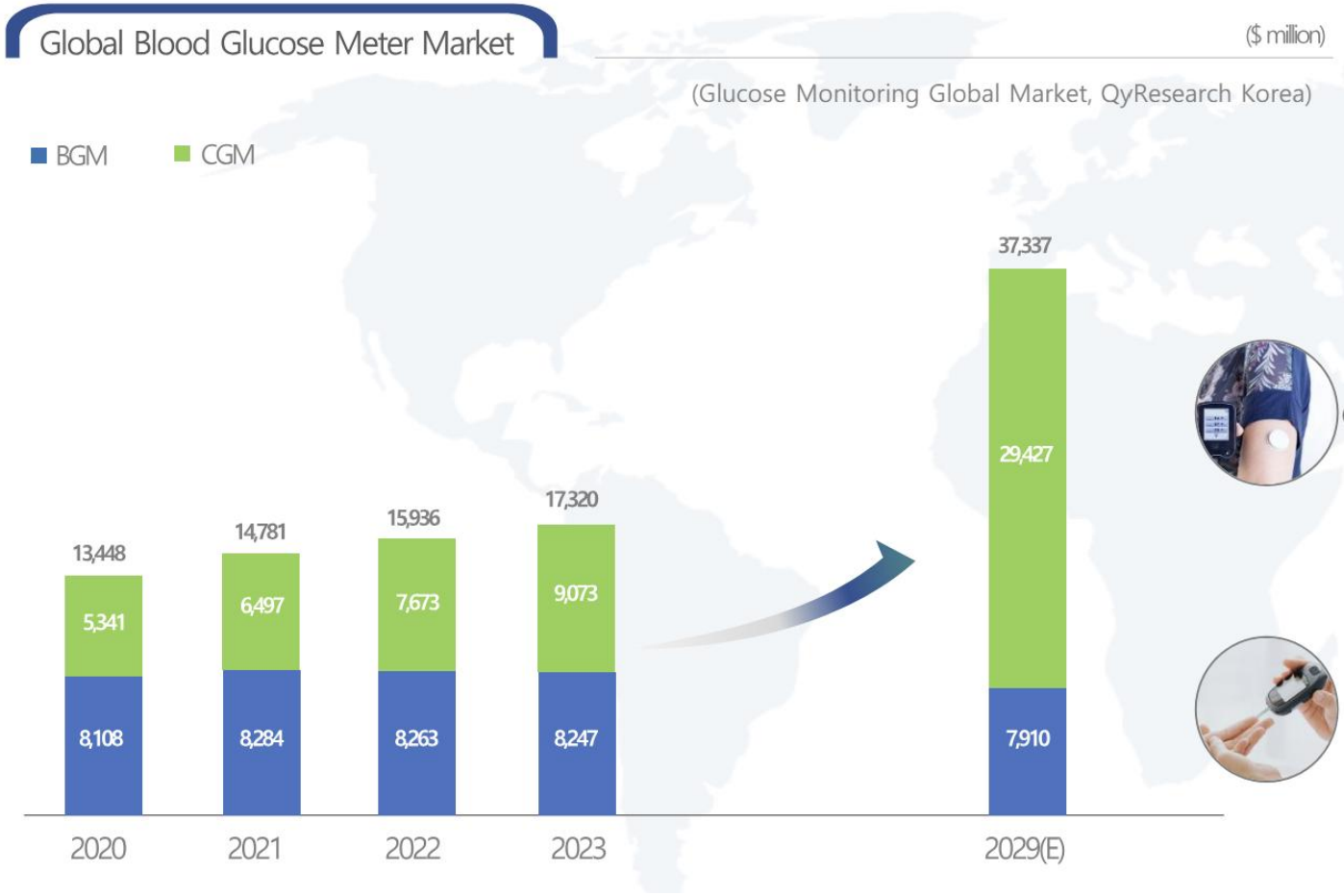
多感科技

2026/01

MIXOSENSE

Confidential

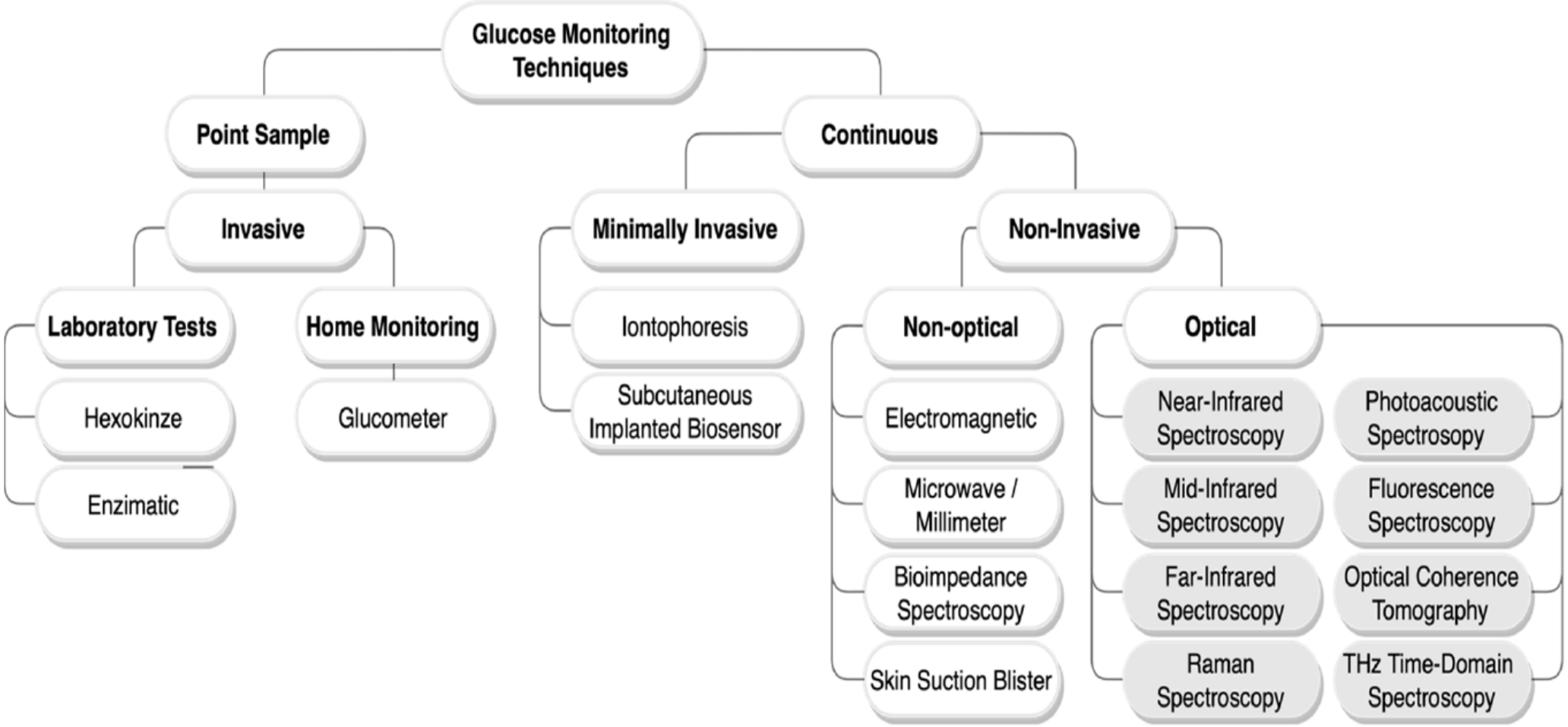
连续血糖检测（CGM）市场



Continuous Glucose Monitoring (CGM) Market Growth ↑

Reduction in the proportion of blood glucose monitors (BGM) ↓

血糖检测方法


Glucose Monitoring Techniques
Point Sample
Continuous
Invasive
Minimally Invasive
Non-Invasive
Laboratory Tests
Home Monitoring
Iontophoresis
Non-optical
Optical
Hexokinze
Glucometer
Subcutaneous Implanted Biosensor
Electromagnetic
Near-Infrared Spectroscopy
Photoacoustic Spectrosopy
Enzimatic
Microwave / Millimeter
Mid-Infrared Spectroscopy
Fluorescence Spectroscopy
Bioimpedance Spectroscopy
Far-Infrared Spectroscopy
Optical Coherence Tomography
Skin Suction Blister
Raman Spectroscopy
THz Time-Domain Spectroscopy

光学无创血糖检测方法

Table 2. Comparison of various non-invasive optical glucose sensing techniques [14,23,27,30,45,46,62,63].

Technology	Wavelength	Selectivity	Measurement Site	Merits	Drawbacks
NIR spectroscopy	750–2500 nm	Good	Ear lobe, finger, forearm, cheek, lip mucosa, oral mucosa, and tongue	- Low-cost - Easy to implement	- Glucose heterogeneous distributions affect accuracy. - Interferences by other chemical compounds
MIR spectroscopy	2500–10,000 nm	Good, superior to NIR	Finger, skin, and oral mucosa	- Quite accurate - Lightweight - Scattering is low	- Poor skin penetration depth - Expensive - High water abortion
FIR spectroscopy	10–1000 μm	Good	ISF	- Scattering is lower than NIR and MIR - Individual daily calibration is not required	- Difficulty in identifying other molecules than water due to strong water absorption
Raman spectroscopy	Visible light	Excellent	Eye, human skin	- Low sensitivity to water and temperature changes - Great specificity - Low-cost	- Lack of stability in the laser wavelength and intensity - Spectrum acquisition takes time
THz-TDS	30 μm to 3 mm	Good	ISF	- Not affected by background noise.	- Long measuring time - Low spatial and depth resolution
Fluorescence	Ultraviolet light, visible light	Excellent	Tears, human skin	- High sensitivity and specificity to glucose concentration - Not affected by light scattering	- Sensitive to changes in pH and oxygen levels - Susceptible to toxicity problems
PA spectroscopy	Ultraviolet light, NIR, and MIR	Good	Finger, forearm, and aqueous humor	- Unsusceptible to water distortion - Not affected by scattered particles	- Low signal-to-noise ratio - Affected by temperature changes, motion, pulsation, and acoustic noise

由于血液中葡萄糖分子振动和分子内部键的旋转，增加了血液的近红外光吸收能力，通过检测葡萄糖分子中含氢基团（O-H和C-H）对近红外光谱780～2500 nm范围内波长的吸收和散射的变化，完成血糖的定量分析。

拉曼光谱

- 拉曼光谱可以观测到皮肤内血糖特征
- 但目前仍面临几个主要问题
  - 系统难以小型化
  - 测试时间过长
  - 是否能达到医疗级精度

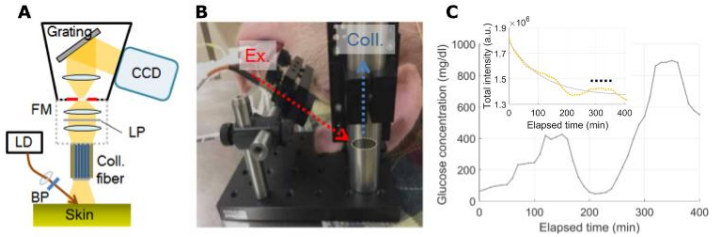


Fig. 5. Raman spectroscopy system, actual probe setup with a subject, and glucose profile during experiment. (**A**) Schematic diagram of Raman spectroscopy system for in vivo animal (swine) skin measurement. FM, *f* number matcher; LP, longpass filter; LD, laser diode; BP, bandpass filter. (**B**) Photograph of Raman probe setup with oblique illumination and vertical collection equipped on top of an anesthetized pig (photo credit: Jeon Woong Kang, MIT). (**C**) Glucose profile during the glucose clamping experiment in trial 1. Exponential time decay of fluorescence from the skin in the inset (spectrum wavelength integration; actual data in the yellow dotted line and its exponential approximation in the gray line). The dotted black line in the inset indicates the time period during which fluorescence stays nearly flat.

Kang *et al.*, *Sci. Adv.* 2020; **6** : eaay5206 24 January 2020 **5 of 8**

三星，MIT

近观科技，瑞金医院

血糖近红外光谱

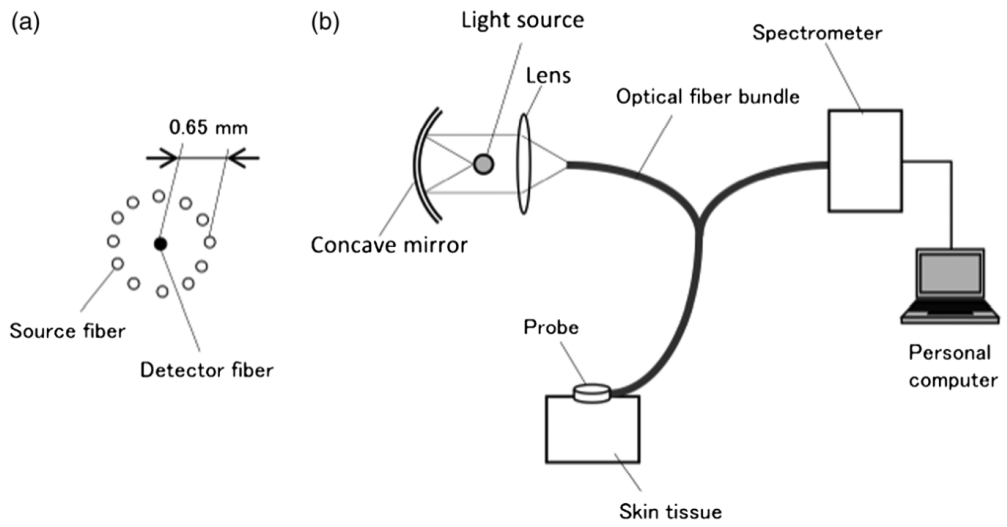


Fig. 2 (a) Arrangement of the source and detector optical fibers at the probe end and (b) schematic diagram of the experimental setup.

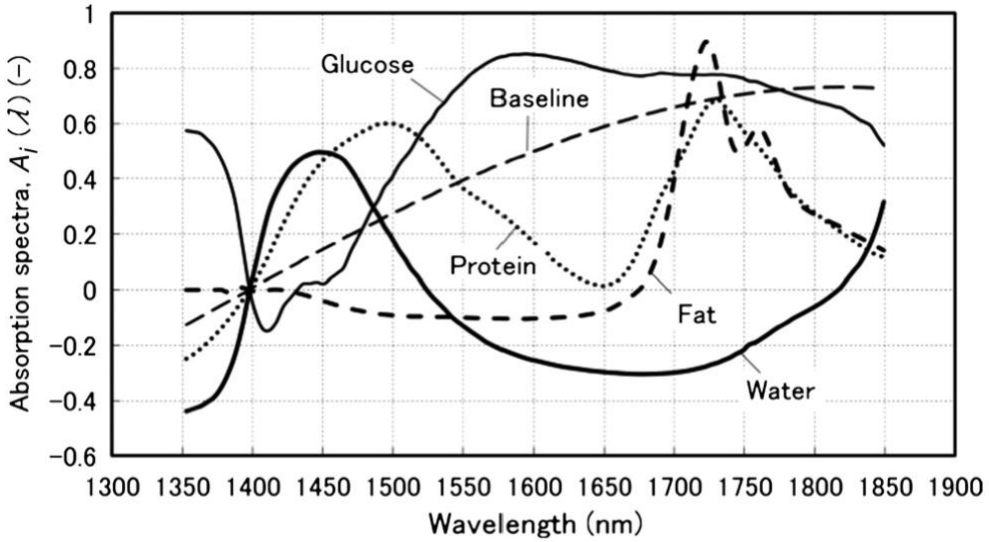
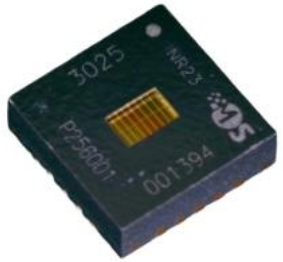
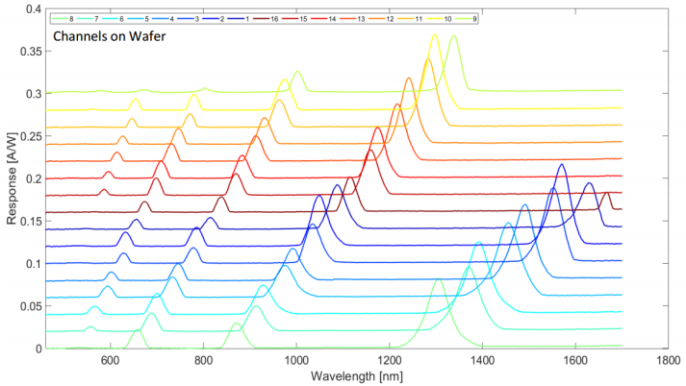


Fig. 1 Absorption spectra, $A_i(\lambda)$, of the four components in living tissues and baseline for synthesizing the absorbance difference spectra.

Maruo K, Yamada Y. Near-infrared noninvasive blood glucose prediction without using multivariate analyses: introduction of imaginary spectra due to scattering change in the skin[J]. Journal of biomedical optics, 2015, 20(4): 047003-047003.

近红外光谱传感解决方案





多通道传感器
MantiSpectra

硅光集成芯片
Rockley Photonics

Topic — Apple

Apple Watch Glucose Monitoring Gets Major Breakthrough

Published January 22, 2026 | Written by TechRepublic Staff

Apple Watch now supports glucose monitoring via CGMs like Dexcom G7, while Apple continues long-term work on non-invasive blood sugar tracking.

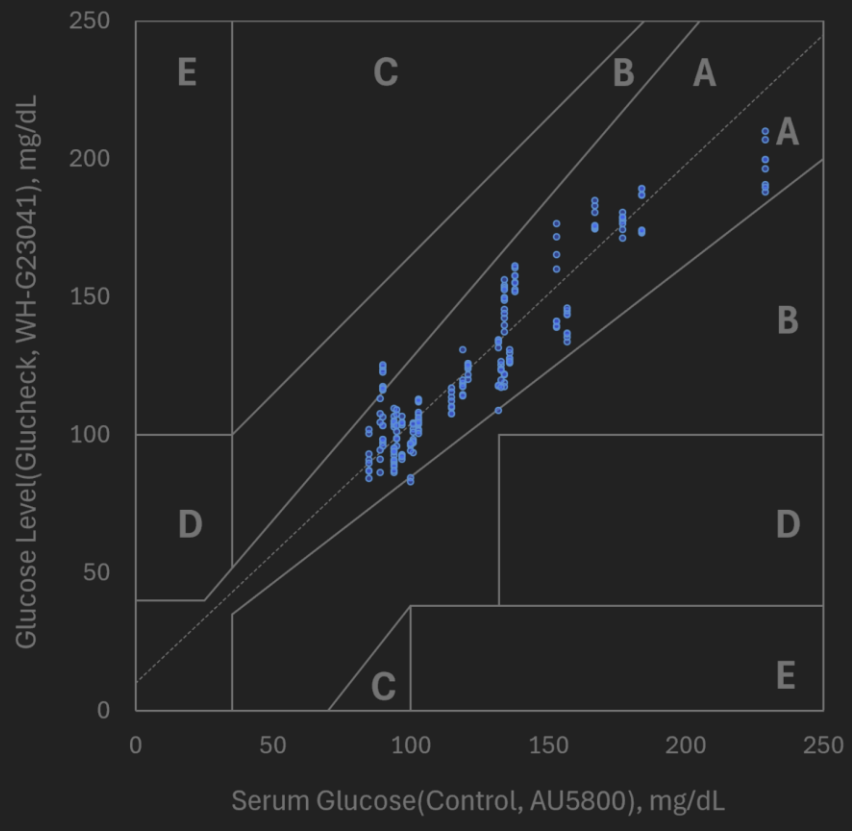
韩国WOORI IO (三星合作)

Pilot Clinical Trial

Location
Korea University Guro Hospital

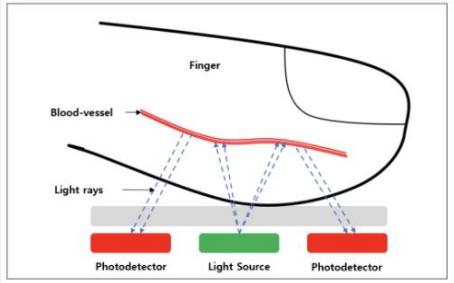
# of Participants
80

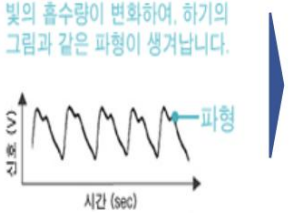
Date
2023.05.09 ~ 2023.09.12

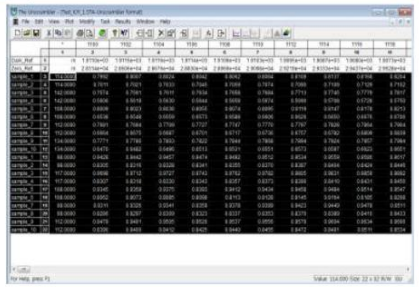


韩国WOORI IO (三星合作)

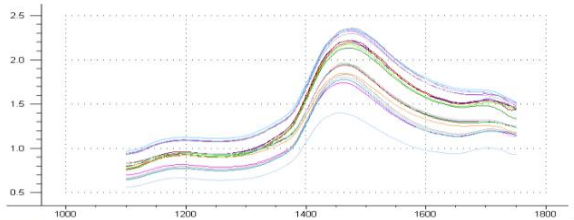
Measurement/Data Collection/Primary Processing

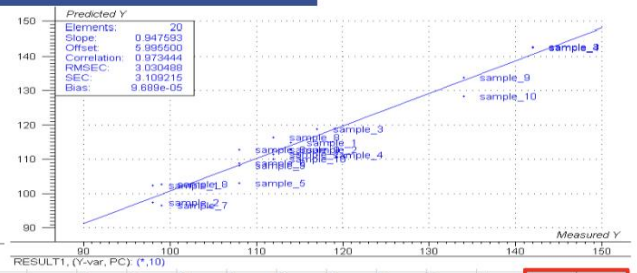






Secondary processing and blood sugar level calculation



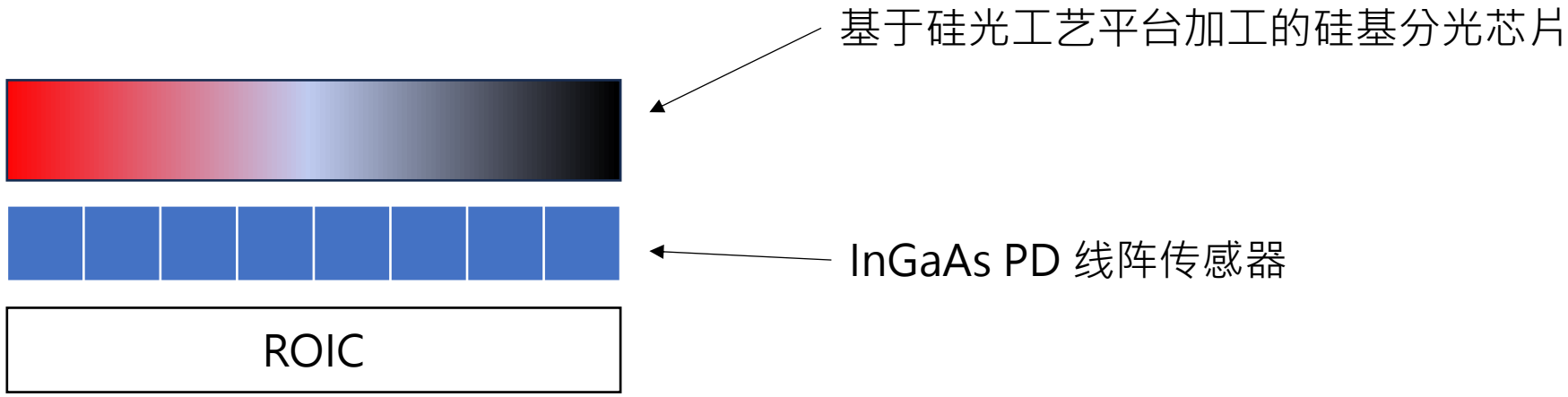


actual	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U		
203	95	17538	17538	17538	17538	16514	14980	12934	10512	8527	7114	6272	6003	6162	5886	5025	3706	2649	1975	1773	1773	210.918	228.134
203	81	17535	17540	17540	17540	16515	14979	12929	10503	8515	7099	6256	5987	6147	5874	5017	3703	2649	1977	1775	1775	194.515	207.948
203	97	17534	17540	17540	17540	16514	14976	12924	10495	8504	7087	6243	5974	6135	5863	5008	3697	2645	1975	1774	1774	195.595	210.302
203	118	17531	17540	17540	17540	16515	14977	12926	10497	8505	7087	6242	5971	6130	5858	5005	3697	2648	1978	1772	1772	193.692	209.232
203	142	17484	17540	17540	17540	16521	14985	12932	10496	8500	7078	6231	5961	6121	5851	5000	3695	2647	1978	1773	1773	196.373	215.198
130	11	17539	17540	17540	17540	16565	15103	13154	10847	8959	7619	6826	6582	6734	6365	5317	3743	2512	1772	1762	1762	118.201	135.391
130	8	17539	17540	17540	17540	16562	15096	13141	10827	8932	7583	6782	6530	6673	6307	5274	3723	2510	1779	1758	1758	130.397	149.157
130	8	17539	17541	17541	17541	16562	15096	13140	10824	8927	7577	6774	6520	6662	6296	5265	3719	2508	1778	1755	1755	124.176	140.172
130	8	17540	17541	17541	17541	16559	15088	13126	10803	8900	7546	6740	6485	6627	6266	5244	3709	2507	1782	1760	1760	127.089	146.96
130	8	17540	17541	17541	17541	16560	15089	13127	10806	8902	7547	6739	6481	6620	6257	5236	3704	2504	1780	1754	1754	129.913	147.418
115	76	17541	17541	17541	17541	16556	15080	13112	10781	8869	7506	6689	6422	6557	6223	5268	3830	2684	1963	1790	1790	109.441	132.419
115	73	17541	17542	17542	17542	16556	15079	13110	10777	8863	7496	6676	6405	6536	6201	5249	3817	2676	1959	1789	1789	105.336	129.254
115	71	17541	17542	17542	17542	16555	15075	13102	10766	8849	7480	6660	6390	6523	6191	5242	3812	2674	1958	1788	1788	106.747	128.984
115	75	17541	17542	17542	17542	16554	15073	13098	10760	8842	7473	6653	6384	6519	6188	5240	3809	2671	1955	1788	1788	107.728	130.145
115	77	17541	17542	17542	17542	16553	15070	13093	10753	8832	7461	6639	6369	6503	6174	5230	3806	2671	1957	1784	1784	111.961	132.04

Data Visualization

1. Apply 7 Light Sources
2. Transflectance Method Application
3. Baseline – Signal Mixed Method applied
4. Apply Monolight Sequence
5. Application of hierarchical separation technique
6. Application of Multi Algorithm based on PLSR

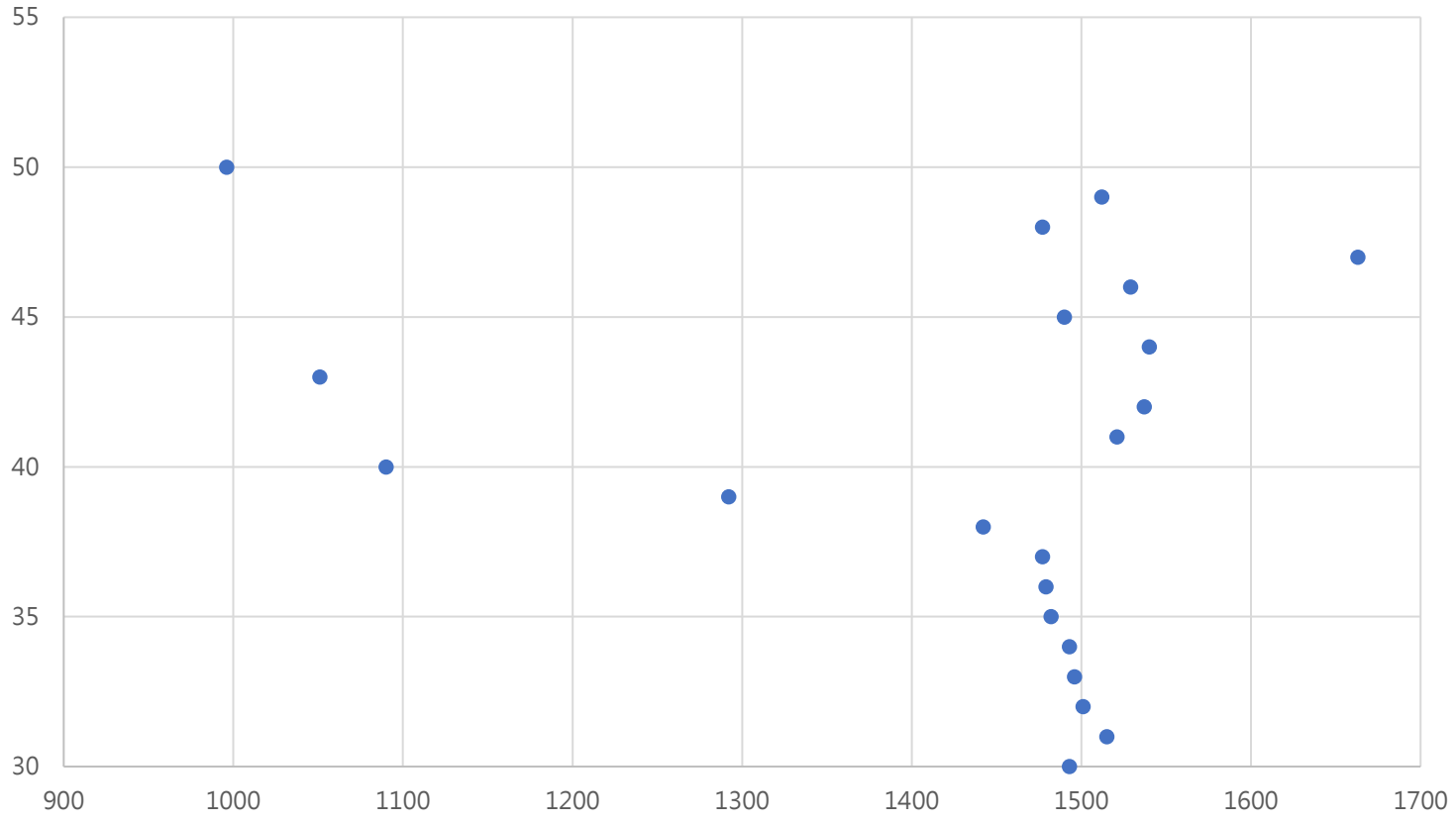
多感科技产品概念



- 近红外波长范围光谱芯片
- 设计目标：小体积、低成本、易集成

动物血糖无创近红外光谱测量初步结果（多感科技与大学合作成果）

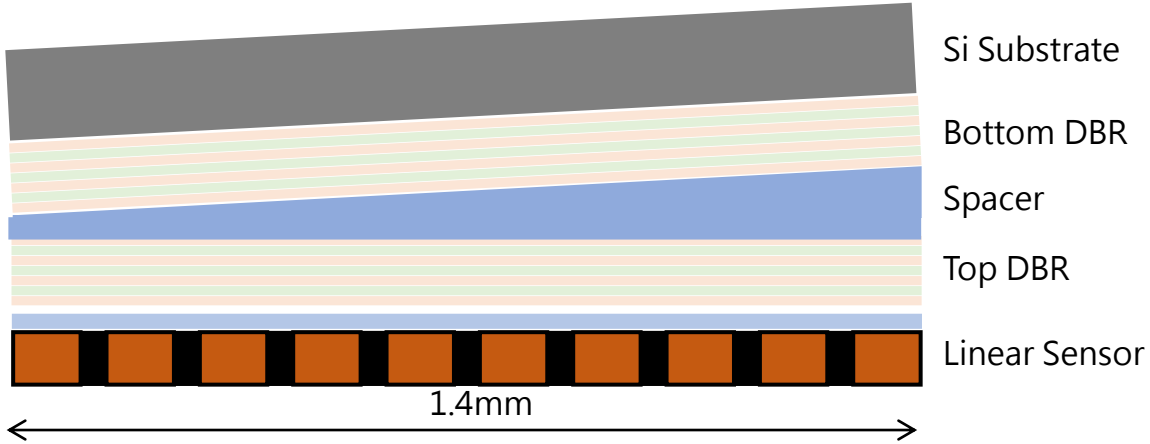
相关性（越高的波长越重要）



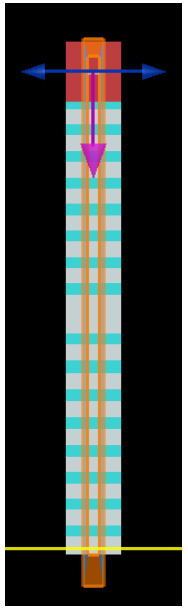
波长（nm）

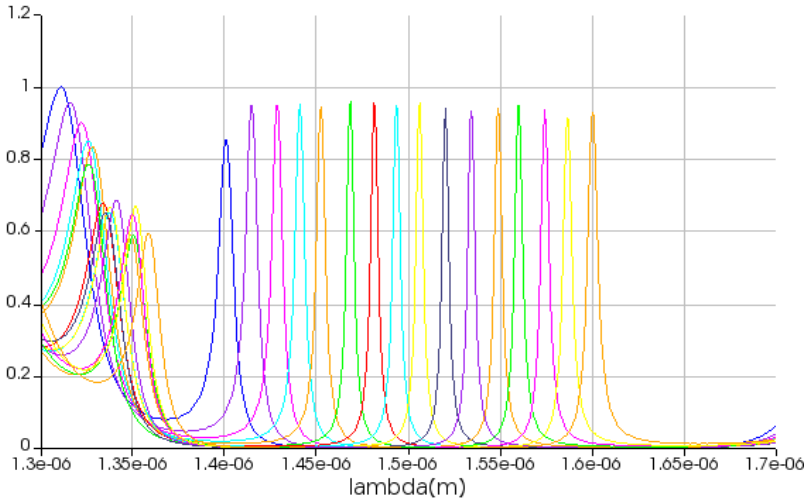
高血糖信息量范围

分光芯片设计细节

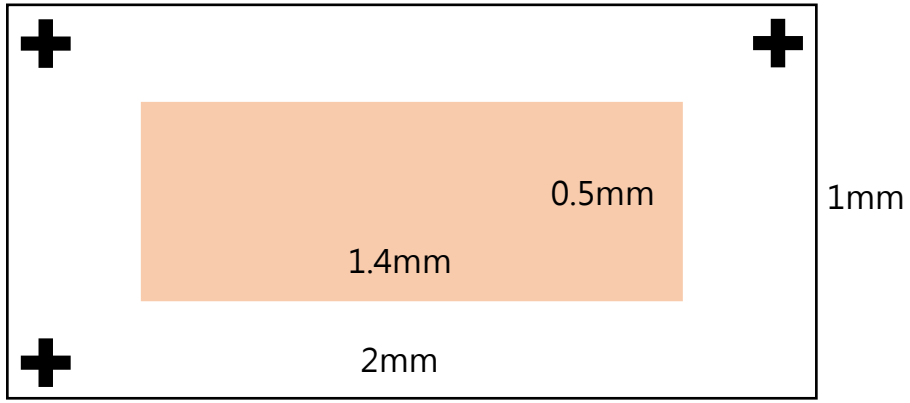


截面示意图





仿真光谱曲线



Top down示意图

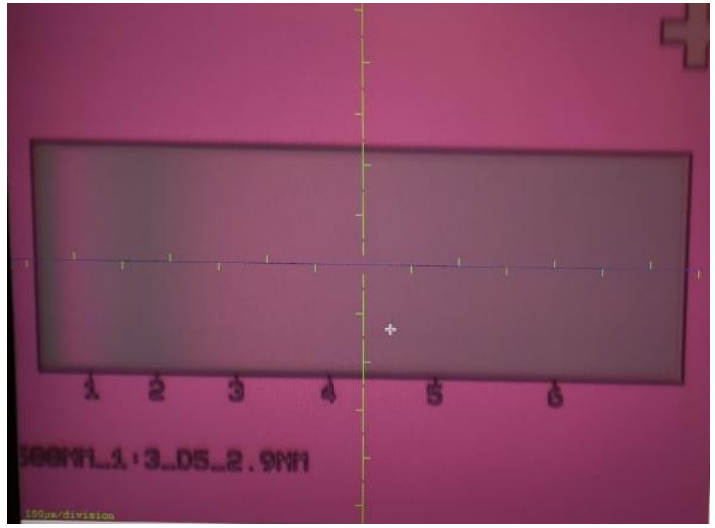
DBR为两种折射率材料堆叠的多层结构

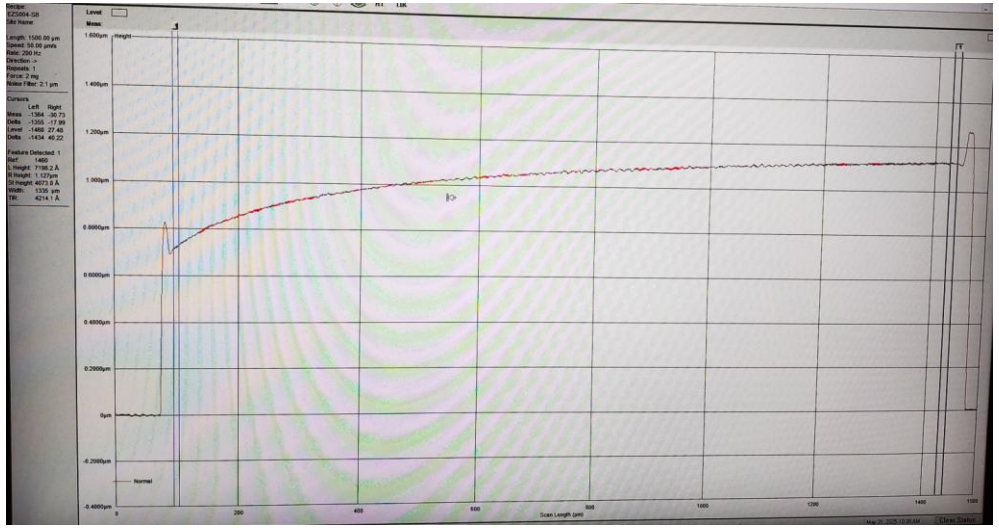
Spacer厚度渐变，通过特殊光刻和刻蚀工艺精确控制

薄膜厚度偏差和均匀性控制在2%以内

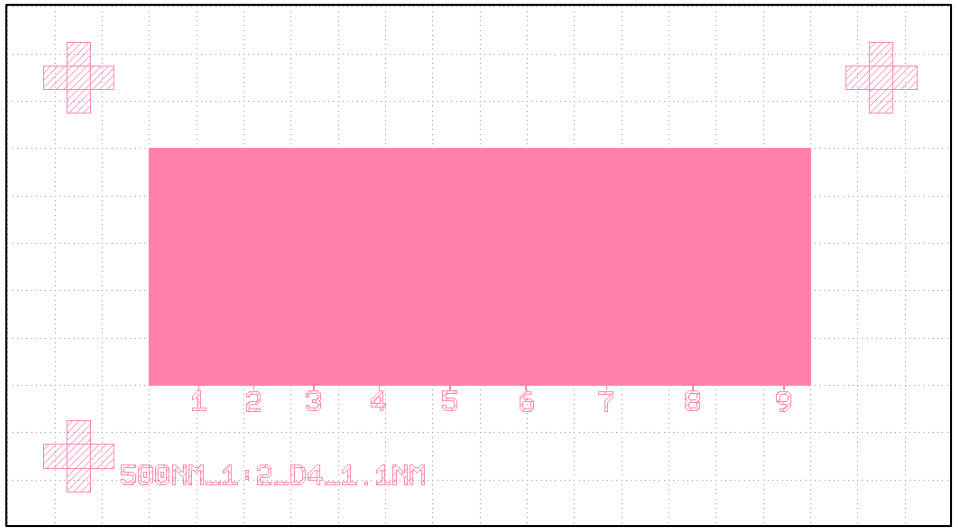
工艺开发结果

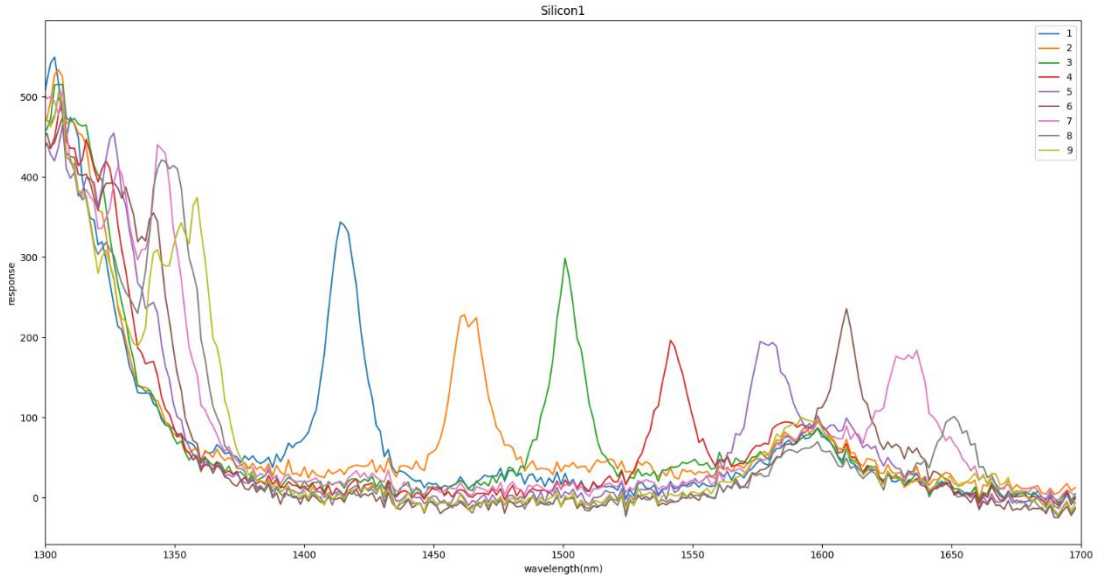
- 开发特殊光刻和刻蚀工艺，实现关键层材料厚度渐变精确控制
- 基于8吋硅光工艺平台开发，易于量产和工艺转移





实测透过光谱数据

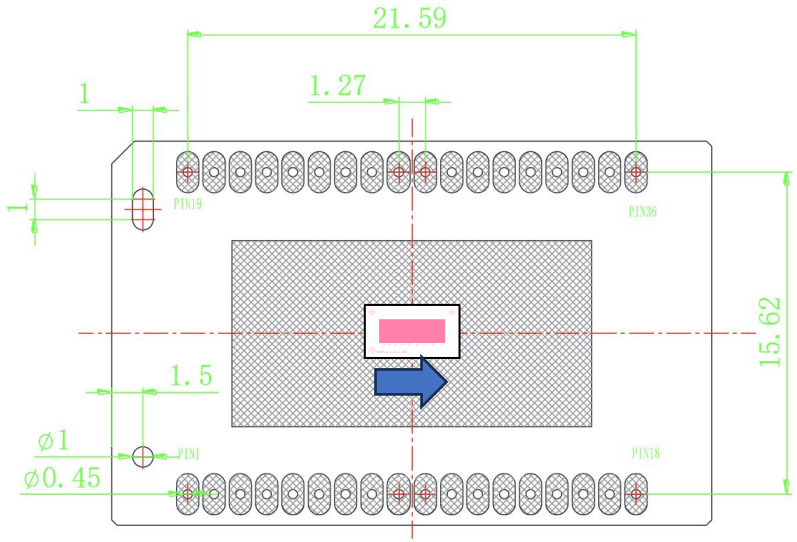


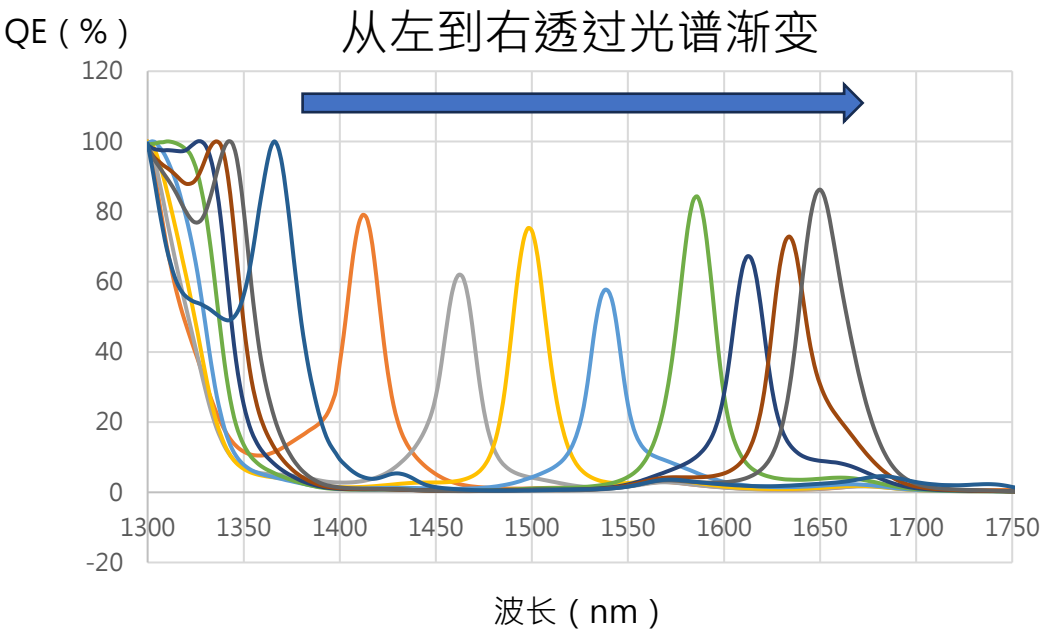


透光区域（0.5x1.4 mm）从左至右，实现精确透过光谱渐变

模组集成与实测QE结果

- 分光芯片集成至12.5 μm像素InGaAs线阵sensor – 112通道近红外多光谱传感器
- Sensor在不同位置像素实测QE符合设计预期





总结和下一步计划

- 完成100+通道超小型近红外光谱传感器原型搭建
- 实测sensor响应效果符合设计预期

- 下一步计划
  - 模组级集成（预计第一版demo样品和2026年Q1完成，Q1底可提供开发板）
  - 和客户和科研院所合作，进行初步数据采集和算法开发
  - 现有方案可以拓展至其他光谱范围